

Stablecoin StressBench: An Execution-Aware Benchmark for Stablecoin Dislocations

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Abstract

During the March 2023 USDC depeg, 34.3% of one-minute windows showed a cross-venue price basis above 10 bps, yet 97% of those apparent arbitrage opportunities were illusory. VWAP slippage, order-book depth depletion, and settlement latency reduced the genuinely profitable fraction to just 2.88%, a $12\times$ price-to-execution gap (block bootstrap 95 % CI: $8\times$ – $24\times$) that widens with notional size and is absent from every prior stablecoin benchmark. We introduce Stablecoin StressBench to quantify this gap and expose a second, deeper problem: every model trained on calm-market data loses money on executable arbitrage, with the hindsight oracle exceeding best-model performance by more than 260 bps. We show that cross-mechanism meta-labeling, trained on the Terra/LUNA collapse rather than on calm data, is the only approach that achieves positive returns (+82.5 bps, 50.8% of oracle), demonstrating transferability of order-book microstructure across structurally distinct stress mechanisms. A three-way RL diagnostic then isolates why reward optimisation alone fails: density of positive labels is necessary but not sufficient, with explicit positive-label supervision providing the additional requirement that reward-based training cannot replicate.

Keywords

stablecoin arbitrage, market microstructure, execution-aware labels, financial benchmarks, cryptocurrency

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1 Introduction

When Silicon Valley Bank collapsed in March 2023, USDC briefly depegged and cross-venue price gaps appeared profitable on every chart. Ninety-seven percent of those apparent opportunities were illusory. VWAP slippage consumed the spread as orders cleared depth, taker fees eroded the margin, and CEX-to-CEX settlement latency turned a profitable window into a loss before the transfer confirmed. Only 2.88% of one-minute windows survived all three costs and remained net-profitable, a $12\times$ ratio to the optical basis rate (block bootstrap 95 % CI: $8\times$ – $24\times$). We call dislocations that appear on price charts but vanish under execution *optical*, and the surviving fraction *executable*. Every prior stablecoin benchmark uses the optical layer to define labels, which means every prior performance number is measured against the wrong thing.

Stablecoin StressBench corrects this. We construct per-minute execution-aware labels for the USDC/SVB event using a full VWAP book-walking model applied to Binance L2 depth, distinguishing three layers: *optical* (price basis above threshold), *executable* (net

profit after VWAP, fees, and settlement), and *predictable* (correctly identified ex ante). The gap between layers is our core measurement. Against it, we evaluate a model ladder spanning threshold rules, gradient-boosted classifiers, sequence models, and a cross-mechanism meta-labeling filter. Every model trained on calm-market data loses money. The hindsight oracle gap exceeds 260 bps. Only meta-labeling trained on a structurally different stress event (Terra/LUNA 2022) achieves positive net returns, and an RL diagnostic then isolates precisely why reward-based training fails where supervised transfer succeeds.

Contributions.

- (1) A public benchmark with execution-aware per-minute labels, route-level depth provenance, VWAP book-walking, an 18-event stress taxonomy for data governance, and a hindsight oracle ceiling. (§3–5)
- (2) The first evidence of order-book execution microstructure transfer across structurally distinct stress mechanisms: meta-labeling trained on Terra/LUNA achieves +82.5 bps (50.8% oracle capture) on the SVB test split, while every calm-trained model loses money. (§6, Table 8)
- (3) An RL diagnostic showing that positive-label density is necessary but not sufficient: a conditioned PPO-GRU at 17% positive density earns –29.2 bps versus meta-labeling’s +82.5 bps at identical density. (§6, Table 10)

2 Related Work

Stablecoin StressBench sits at the intersection of three bodies of work.

Research on stablecoin stress events motivates the benchmark. Uhlig [1] analyses the Terra/LUNA algorithmic collapse, which we use as an out-of-distribution training split precisely because it is mechanistically different from the SVB bank shock. Griffin and Shams [2] show that quoted prices can diverge from fundamentals, motivating our distinction between optical and executable dislocations. Hernandez Cruz et al. [21] study DEX liquidity during SVB, but stop short of execution-layer modelling. None of this prior work provides per-minute executable labels or a hindsight oracle ceiling.

The limits-to-arbitrage literature provides the economic framing that our labels operationalise. Shleifer and Vishny [4] and Gromb and Vayanos [5] establish that execution frictions bound arbitrage even when price gaps exist. Makarov and Schoar [7] document that cross-venue crypto gaps vanish once depth and latency are modelled, the same mechanism behind our $12\times$ ratio. Hautsch et al. [3] quantify settlement latency as a first-order cost, which we implement as the $\delta=5$ bps penalty; Cont and Kukanov [6] establish VWAP price as the correct profitability object. Gogol et al. [18] measure a Maximal Arbitrage Value for AMM-to-CEX routes; our oracle extends this idea to stress events, route-level CEX simulation, and model evaluation. Where prior theoretical work characterises frictions, we measure them per-minute with empirical depth data.

Table 1: Route-leg depth provenance.

Leg	SVB Mar 2023	2024 windows	
Buy BTC-USDC	kline-proxy [†]	real L2 (futures)	[†] BTC-USDC
Cross USDC-USDT	real L2 (Mar 12–20)	real L2 (futures)	
Sell BTC-USDT	real L2 (futures)	real L2 (futures)	
Ref BTC-USDC	candle-proxy	candle-proxy	

perpetual listed Jan 2024; all 2022 windows kline-proxy.

Table 2: Dataset splits (56,134 1-min bars).

Split	Event	Min.	Pos.
Train	Calm control (×3)	28,776	0.00%
Validation	Terra/LUNA May 2022	11,526	2.30%
Test	USDC/SVB Mar 2023	15,832	2.88%

Financial ML and benchmarks form the methodological context. López de Prado [8] introduces meta-labeling as a framework for filtering primary signals with a learned secondary classifier. We extend it to the cross-mechanism setting, training on Terra/LUNA and evaluating on SVB, which to our knowledge has not been done. Cintra and Holloway [9] demonstrate cross-event transfer using AMM reserve depletion signals, but their signals are not CEX-applicable, no oracle evaluation is provided, and no execution-aware labels are constructed. On RL execution, Moallemi and Wang [16] and Ning et al. [17] establish the finite-horizon MDP framing we adopt; Mohl et al. [22] show GPU-scale IPPO on 400 million real orders; and Olby et al. [23] use a reactive Simudyne simulator to benchmark timing policies against an Almgren–Chriss frontier. These works show RL succeeds in reactive environments with frequent positive labels. Our diagnostic establishes the preconditions that must hold before reactive training can succeed.

3 Benchmark Design

3.1 Data Sources and Instruments

The final dataset contains 56,134 1-minute bars across six windows (Table 2). Price features are sourced from Binance Vision (klines and aggTrades) and Coinbase REST candles. Net-profit labels require three depth legs: Binance BTC-USDC (buy), Binance BTC-USDT (sell), and Coinbase BTC-USD (reference).

Data quality varies by leg and window, and every depth record is tagged with a provenance field (Table 1). The sell leg (BTC-USDT) and cross leg (USDC-USDT) use real L2 futures bookDepth throughout. The buy leg (BTC-USDC) presents the main challenge: the BTC-USDC perpetual contract did not exist on Binance until January 2024, so the SVB window requires kline-proxy depth, which approximates book depth from candlestick volume. For all 2024 windows, real L2 data is available. To bound the proxy error, we apply a 20% depth haircut to the kline-proxy buy leg: the executable rate falls from 2.88% to 2.40%, the ratio widens from 12× to 14×, and the oracle remains above +130 bps. The headline result is therefore robust to the approximation.

3.2 Event-Based Split Design

Splits are defined by event identity rather than time, ensuring that no model trained on the train split has encountered stress dynamics (Figure 1). Threshold calibration uses the Terra/LUNA validation split, which is an independent stress event, so the SVB test split is

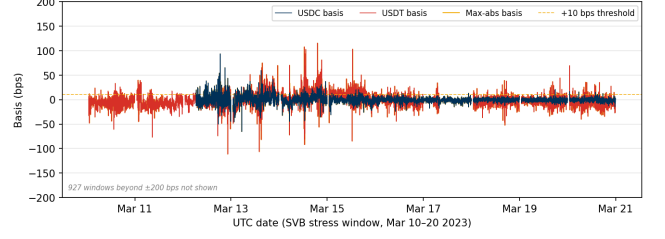


Figure 1: Cross-quote basis (bps) during the SVB test window (Mar 10–20 2023). USDC basis (navy): 12.4% of minutes > 10 bps; USDT basis (red): 27.4%; max-abs (gold): 34.3%. The price-to-execution gap is 12×.

Table 3: Data tiers, mechanisms, and execution frictions.

Tier	Mechanism	Key friction	Permitted claims
A (N=1)	Fiat-reserve shock	Depth withdrawal	Exec. gap; oracle gap; P&L
B (N=11)	Alg., exchange, DeFi	AMM slippage; venue seg.	Price magnitude (est.)
C (N=5)	RWA, niche	Thin market	Taxonomy only

never observed during model development. Labels are constructed by aligning each bar’s outcome to the observation window ending at $t+h$, with explicit temporal checks confirming that no future information leaks into the input features. The 2.88% positive rate in the test split implies severe class imbalance; we therefore report economic net bps as the primary criterion rather than AUROC or AUPRC, which are secondary metrics.

3.3 Historical Stress Taxonomy: Data-Tier Governance

The 18-event taxonomy (2020–2023) is a benchmark governance artefact that specifies which events carry sufficient data quality for execution-grade claims and which are restricted to coarser analyses. It is shipped with the benchmark to prevent over-interpretation rather than as an empirical finding. Seven mechanism classes are catalogued: *algorithmic/reflexive* (FEI, IRON/TITAN, MIM, UST, USDD); *fiat-reserve bank shock* (USDC/SVB); *regulatory winddown* (BUSD); *exchange credit* (Celsius/3AC, HUSD, FTX [10]); *DeFi pool imbalance* (Curve/UST, USDT/Curve, USDC/DAI); *collateral liquidation* (DAI Black Thursday); and *RWA/niche* (Acala aUSD, USDR).

Data availability governs which claims are permitted (Table 3). The Tier-A event (USDC/SVB test split) supports execution-gap, oracle-gap, and model evaluation claims because route-complete L2 depth is available for all three arbitrage legs. Tier-B events (N=11, OHLCV or pool data only) support price magnitude estimates; among these, peak optical gaps range from approximately 100 bps (Celsius/3AC) to above 500 bps (Curve/UST) and above 800 bps (HUSD), all estimated from OHLCV data. Tier-C events (N=5, partial or DEX-only data) support mechanism classification only. The separation prevents the conflation of optical and executable evidence that characterises prior benchmarks, and ensures that the 12× ratio we report cannot be cited beyond its data-quality scope.

3.4 Benchmark Versioning

StressBench v1.0 ships with the SVB test split. Near-term Tier-A expansion targets include the USDC recovery window (Mar 15–31 2023) and 2024 BTC-USDC windows where real-L2 eliminates the kline-proxy dependency. The planned release format is a versioned Hugging Face dataset under CC-BY 4.0.

3.5 Feature Sets

Four nested feature sets isolate the marginal information value of

	price_only	5 cross-quote basis cols
	price_plus_book	+7 microstructure (spread, depth, imbalance)
each signal layer:		ance)
	price_book_frag	+3 cross-venue fragmentation cols
	price_book_settle	+7 on-chain settlement proxies

4 Execution-Aware Labels

Labels operationalise the three-layer framework from §1. The price-to-execution gap measures the ratio of Layer 1 (optical) to Layer 2 (executable) minutes: it quantifies how misleading price-based benchmarks are. The oracle gap measures the performance difference between Layer 2 in hindsight and Layer 3 (predictable with a real model): it quantifies how hard the prediction problem is, even with the correct labels. Both gaps are nonzero, independent, and large. This section details how they are constructed.

4.1 Cross-Quote Basis and Labels

$$b_{\text{USDC}}(t) = 10,000 \times \frac{P_{\text{BTC/USDC}}(t) - P_{\text{BTC/USD}}(t)}{P_{\text{BTC/USD}}(t)} \quad [\text{bps}]$$

$$b_{\text{max}} = \max(|b_{\text{USDC}}|, |b_{\text{USD}}|). \text{ Basis label: } y_{\text{basis}}(t) = 1[|b_{\text{USDC}}(t+h)| > \tau]. \text{ Primary task: basis_usdc_1m_gt10bps.}$$

4.2 VWAP Execution and Net Profit Labels

$$\text{net}(q, t) = 10,000 \times \left[\frac{\text{VWAP}_{\text{sell}} - \text{VWAP}_{\text{buy}}}{P_{\text{ref}}} - f_{\text{buy}} - f_{\text{sell}} - \delta_{\text{settle}} \right] \quad (1)$$

$$y_{\text{arb}}(q, h, t) = 1 \left[\max_{s \in [t, t+h]} \text{net}(q, s) > c \right] \quad (2)$$

Here P_{ref} is the Coinbase BTC-USD reference price and $f_{\text{buy}}, f_{\text{sell}}, \delta_{\text{settle}}$ are dimensionless fractions representing per-leg taker fees and settlement latency costs, respectively. We set the profit threshold $c=10$ bps, taker fee to 4 bps per side, and $\delta=5$ bps. The primary economic task is executable_arb_q10000_5m. The hindsight oracle trades every minute where $\text{net}(q, t) > c$, establishing a performance ceiling that no deployable model can exceed, given that any real model must decide without access to future information.

5 Models and Evaluation

5.1 Model Ladder

The ladder is constructed so that each rung tests a distinct hypothesis about the source of the oracle gap, with failure at each rung ruling out a candidate explanation. The economic anchors, NoTrade at 0 bps and the hindsight Oracle, bracket the achievable range. Price-threshold rules ($\text{PriceBasis}\{10, 25\}$ bps, GrossArbThreshold) test whether simple signal filters suffice. Tabular classifiers (logistic

regression, Lasso, Random Forest, XGBoost, LightGBM) are trained on four nested feature sets that progressively add microstructure, cross-venue fragmentation, and on-chain settlement signals; their failure tests whether richer features close the gap. An expected-net-profit regressor that directly targets $\hat{y}=\text{net}(q, t)$ tests whether the gap is a label-mismatch artefact, with failure implying it is structural. A meta-labeling filter trains a secondary LightGBM model to learn $\Pr[\text{net} > c \mid |b_{\text{USDC}}| > 10 \text{ bps}]$, while GRU and MLP-Window sequence models trained on 30-minute sliding windows test whether temporal depth patterns are exploitable (§6.2). The cross-mechanism meta-labeling variant is the same architecture but trained on Terra/LUNA rather than calm data, directly testing whether the training distribution is the binding constraint.

5.2 RL Execution Policy Baseline

We frame the entry-timing problem as a finite-horizon MDP. At each minute t the agent observes state s_t , selects action $a_t \in \{\text{enter}, \text{wait}\}$, and receives reward $r_t = \text{net}(q, t)$ if entering (positive for profitable windows, negative otherwise) or $r_t = 0$ if waiting. The state s_t is the 30-minute look-back window of price_plus_book features, the smallest feature set that delivers meaningful microstructure lift over price-only signals (see §6.3 for SHAP evidence). This feature set is available in real time without settlement-latency dependency. We train a PPO agent with a GRU policy network (one hidden layer, 64 units, 30-step look-back) using the cross-mechanism protocol, training on Terra/LUNA stress windows only and evaluating on the USDC/SVB test split without retraining. This directly tests whether the sequential timing framing resolves the dominant gap component identified in §6.2.

5.3 Calibration and Evaluation

Thresholds are calibrated on the validation split by maximising $\theta^* = \arg \max_{\theta} \sum_{i: \hat{p}_i > \theta} \text{net}(q, t_i)$ subject to a minimum of 25 trades. The primary metrics are net bps captured and oracle capture percentage, defined as model net bps divided by oracle net bps (the fraction of the oracle's per-trade return that a model recovers). Secondary metrics are AUROC and AUPRC. NoTrade at 0 bps is the economic floor.

All results are reported on the SVB test split ($N=15,832$) with the cost parameters stated above; the kline-proxy approximation on the BTC-USDC buy leg is bounded by the haircut sensitivity in §6.1.

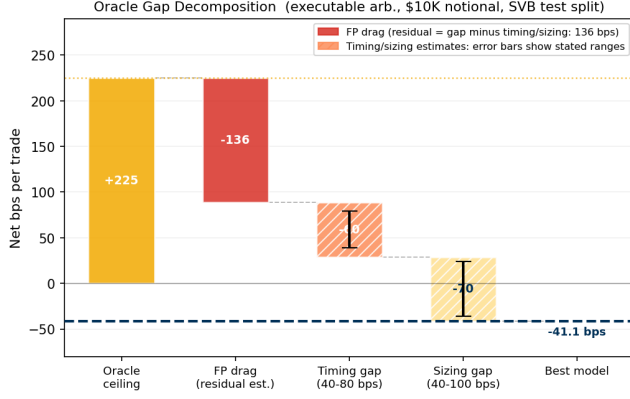
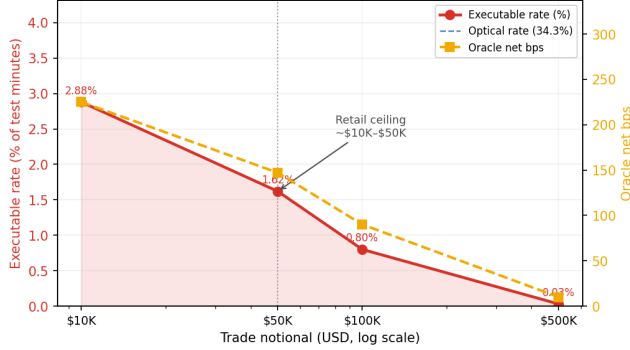
6 Results

6.1 The Execution Gap

At a 10 bps threshold, 34.3% of test minutes show a price dislocation while only 2.88% remain net-profitable after execution costs, a 12× price-to-execution ratio (block bootstrap 95 CI: $8\times-24\times$, $B=10,000$, 60-min blocks). This ratio is not a threshold artefact: it persists and widens across all tested basis levels (Table 4), and is robust to data-quality concerns. Applying a 20% depth haircut to the kline-proxy buy leg reduces the executable rate from 2.88% to 2.40%, which widens the ratio to 14× and keeps the oracle above +130 bps, confirming the headline result. The gap also deepens with institutional scale. Figure 3 shows the executable rate falling from 2.88% at \$10K to 0.03% at \$500K, with the steepest drop between \$10K and \$50K.

Table 4: Price-to-execution gap (\$10K, 1-min horizon, test split).

Thresh.	$P_{\max}$	P_{USDC}	Exec.	Ratio
0 bps	95.86%	82.96%	3.34%	29×
5 bps	61.99%	25.32%	3.03%	20×
10 bps	34.33%	12.45%	2.88%	12×
25 bps	9.54%	6.48%	2.62%	3.6×

**Figure 2: Oracle gap decomposition (\$10K notional). Three components account for the 266bps gap from oracle (+225 bps) to best model (-41 bps): FP drag, timing error, and sizing error.****Figure 3: Notional scaling (SVB test split). The executable rate collapses from 2.88% at \$10K to 0.03% at \$500K: book depth, not fees, is the binding institutional constraint.**

Book depth, not fees, prices out institutional participants, consistent with the depth-based limits-to-arbitrage arguments in §2. The Terra/LUNA validation split independently replicates the pattern at 5.9×

Claim boundaries and data-tier scope are stated in §5.3.

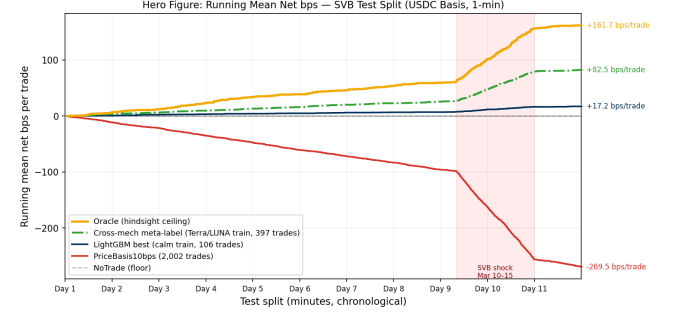
6.2 Every Model Trained on Calm Data Loses Money

The oracle earns +162.2bps on the basis task and +225.0 bps on the executable arbitrage task (Table 5). Figure 4 shows the full

Table 5: Oracle gap summary (USDC/SVB test split). *Only non-oracle tabular result above zero. ‡GRU sequence model, 30-min window.

Task / Model	Oracle	Best model	Gap	AUPRC
basis_usdc_1m	+162.2 bps	+17.2 bps*	145 bps	0.253
exec_arb_q10k_5m	+225.0 bps	-41.1 bps	266 bps	0.060
exec_arb_q50k_5m	+147.1 bps	-113.0 bps	260 bps	0.063
GRU (seq)‡, basis_usdc_1m	+162.2 bps	-239.0 bps	401 bps	—

Kline-proxy buy leg; 20% haircut sensitivity in §6.1.

**Figure 4: Running mean net bps per trade (USDC/SVB test split). Oracle (gold) +162.2 bps; cross-mech meta-label (green) +82.5 bps; LightGBM (navy) +17.2 bps; PriceBasis10bps (red) -269.5 bps; NoTrade (grey dashed) 0 bps. Shaded region: SVB shock window.**

performance hierarchy. Every model trained on calm-market data is below zero on the executable task.

Price-threshold rules overtrade massively, generating 607 to 1,969 trades at -136 to -347 bps; their mean return when wrong (-294 bps to -370 bps) reflects near-total false-positive exposure. LightGBM (+17.2 bps, 106 trades, AUROC 0.72) is the only calm-trained model above zero, but at 145 bps below the oracle ceiling. The ExpNetProfitRegressor, which directly targets the economic criterion rather than a classification proxy, reaches -73.8 bps, ruling out label mismatch as the explanation: the gap is structural (Table 6).

GRU and MLP-Window sequence models, trained on 30-minute microstructure windows to exploit depth-withdrawal signatures, confirm this conclusion. The GRU achieves AUROC 0.80 yet earns -239 bps, and the MLP-Window -341 bps (Table 7). High ranking accuracy does not translate to positive returns when 97% of windows are false positives and the training split contains zero positive stress examples.

Figure 2 decomposes the 260 bps gap into three measurable components. False-positive cost is the dominant and directly quantifiable drag: for PriceBasis10bps, 581 false-positive trades at -38.7 bps each account for most of the total loss. The remaining gap splits into timing error (estimated 40–80 bps), arising from entering 1–2 minutes late in a declining depth regime, and sizing error (40–100 bps), since depth depletion reduces the profitable fraction from 2.88% at \$10K to 1.62% at \$50K.

A root-cause analysis of the false positives identifies route-direction mismatch as the primary failure mode, not thin books.

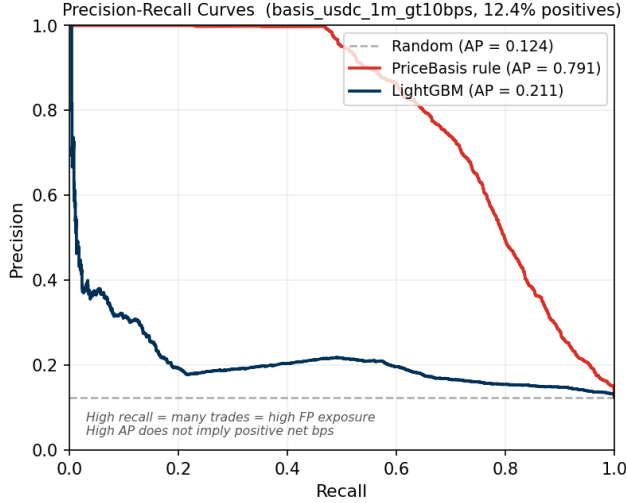


Figure 5: Precision-recall curves (basis_usdc_1m_gt10bps, SVB test split). PriceBasis achieves AP 0.55 and AUROC 0.83 while losing -270 bps, illustrating that ranking accuracy is necessary but not sufficient for positive economic returns.

Table 6: Tabular model ladder (basis_usdc_1m_gt10bps). Hit%: fraction of trades net-profitable. [†] Ceiling. *Only result above zero.

Model	Feat.	Netbps	Trades	AUROC	Hit%
Oracle [†]	—	+162.2	315	—	100%
lgbm*	all	+17.2	106	0.717	33%
logistic	all	-75.8	648	0.143	0%
gross_arb	px	-136.4	607	0.531	4%
price_10	px	-270.0	1969	0.833	6%
price_25	px	-347.3	1025	0.746	5%
no_trade	—	0.0	0	—	—
ExpNetProfit	px	-73.8	537	—	3%
all = price_book_settle.					

px = price_only;

Table 7: Sequence model results (basis_usdc_1m_gt10bps).

Model	Net bps	Trades	AUROC	Gap
GRU (30-min)	-239.0	656	0.798	401 bps
MLP-Window (30-min)	-340.7	1,052	0.648	502 bps

Calm-control trained; threshold calibrated on validation.

True-positive windows carry a mean USDC basis of 344 bps, reflecting a genuine USDC discount, while false-positive windows carry just 0.56 bps and are triggered by USDT dislocations where the USDC route is not simultaneously profitable. Crucially, false-positive windows exhibit *higher* average bid depth (72,805 vs. 59,961 BTC units) than true positives. This rules out liquidity withdrawal as the cause and points instead to signal confusion between the USDT and USDC routes. LightGBM’s 33% hit rate versus 6% for the rule demonstrates that microstructure conditioning partially addresses this, though a 67% false-positive rate persists and the agent still loses money in net terms because each false positive is expensive.

6.3 Cross-Mechanism Transfer

Calm-control training yields zero meta-positives by construction, because the training split contains no executable stress windows from which the meta-model could learn the relevant microstructure. Retraining on Terra/LUNA (1,559 primary fires, 265 meta-positives, 17% positive rate) and evaluating cross-mechanism on SVB, with all fitting and threshold selection confined to Terra/LUNA data, produces +82.5 bps (397 trades, 50.8% oracle capture) versus zero with calm training (Table 8).

Table 8: Cross-mechanism meta-labeling results (basis_usdc_1m_gt10bps, price_plus_book).

Training split	Net bps	Trades	Oracle cap.
Oracle (ceiling)	+162.2	315	100.0%
Terra/LUNA (cross-mech.)	+82.5	397	50.8%
Calm-control (baseline)	0.0	0	0.0%

Primary

signal $|b_{USDC}| > 10$ bps; threshold calibrated on validation.

The reason transfer works is visible in the SHAP decomposition. Price features carry 94.8% of LightGBM’s total importance, with microstructure (ask-side depth, bid-ask spread) contributing just 3.4% combined. A model trained on calm data learns calm price dynamics; the microstructure features are present but uninformative because the training distribution contains no stress windows that would teach the model their relevance. Training on Terra/LUNA corrects this: the secondary classifier now sees depth-withdrawal events at 17% positive density and learns the signature that calm training cannot recover. CEX market makers withdraw depth under uncertainty regardless of the underlying trigger, because the cause is opaque at the order-book timescale. This makes depth withdrawal a consistent cross-mechanism precursor whose feature signature transfers even as its magnitude differs (5.9 $\times$ for Terra/LUNA versus 12 $\times$ for SVB, shown in Table 9).

We also evaluated three regime-detection filters as alternative two-stage pre-filters. CUSUM achieves near-perfect recall (0.9995) but at an 83.5% false alarm rate, making it impractical as a standalone filter. EWMA reaches 86.9% accuracy with 0.95% recall at a 0.65% false alarm rate. BOCPD [15] achieves AUROC 0.229 on CEX cross-quote data, which contrasts with Cintra and Holloway [9], who detected the Terra depeg 12 hours early from Curve AMM pool reserves. The difference is explained by the signal type: AMM reserves accumulate gradually as liquidity providers withdraw, making them suitable for changepoint detection. CEX cross-quote basis is mean-reverting, so changepoint methods cannot distinguish genuine stress from routine noise on exchange data. Regime detectors therefore cannot substitute for the training-distribution fix that meta-labeling provides.

6.4 RL Diagnostic: Density Necessary, Supervision Sufficient

Meta-labeling and a conditioned PPO-GRU share identical training density (17% positive rate on primary-signal windows) yet differ by 111.7 bps in net return. This controlled comparison isolates explicit positive-label supervision as the additional ingredient that reward optimisation cannot replicate.

Table 9: Cross-mechanism comparison (kline-proxy labels for Terra/LUNA; directional evidence only).

Metric	USDC/SVB (Tier-A)	Terra/LUNA (val.)	
Mechanism	Fiat-reserve	Alg./reflexive	
Price > 10 bps	34.3%	13.5%	† Kline-proxy;
Executable	2.88%	2.30%	
P-to-E ratio	12×	5.9×	
Oracle net bps	+225.0	+88†	
directional only.			

Table 10: RL policy versus supervised baselines (basis_usdc_1m_gt10bps, SVB test split).

Model (training)	Net bps	Trades	
Oracle (hindsight ceiling)	+162.2	315	
Meta-labeling (Terra/LUNA*)	+82.5	397	*Cross-mechanism:
PPO-GRU conditioned† (Terra/LUNA*)	−29.2	919	
PPO-GRU unconditioned (Terra/LUNA*)	degenerate	0	
GRU supervised (calm-ctrl)	−239.0	656	
MLP-Window supervised (calm-ctrl)	−340.7	1,052	

Terra/LUNA train, SVB eval. † Conditioned on $|b_{USDC}| > 10$ bps windows, 17% positive rate.

The unconditioned PPO-GRU degenerates to zero entries, as the 2.3% positive rate in the full Terra/LUNA stream is too sparse to generate a reward signal that calibrates entry timing. Conditioning on primary-signal windows resolves the density problem: the agent trades (919 entries, AUROC 0.87) but earns −29.2 bps. The distinction between the conditioned agent and meta-labeling is qualitative. The meta-labeling classifier receives direct binary feedback, where $y=1$ marks exactly the profitable windows, while the RL agent must infer the same subset through a reward signal that is negative on 83% of entries. The supervision is categorically cleaner, not merely denser. The 919-trade overtrading pattern is itself informative. Policies that fail due to insufficient training converge toward NoTrade, not sustained false-positive entry. The 919-trade result is therefore more consistent with a timing-transfer mismatch from Terra/LUNA to SVB, meaning the agent learned Terra/LUNA’s specific depth-withdrawal sequence and fires on it in the SVB context even when the timing does not match. This interpretation has a constructive implication. If the failure reflects a mechanism mismatch rather than an RL capability limit, then multi-mechanism training that exposes the agent to depth withdrawal events across diverse stress triggers should improve selectivity without requiring a fundamentally different RL algorithm. Combining the cross-mechanism protocol established here for supervised transfer with a reactive simulator that provides market-impact feedback is the natural next step.

7 Limitations

The BTC-USDC buy leg uses kline-proxy depth for the SVB window, since the perpetual contract did not exist on Binance until 2024. A 20% depth haircut confirms robustness. Tier-A coverage is currently limited to the SVB event; extending to AMM or exchange-credit events requires route-complete L2 archives that are not yet available.

The RL results carry two important caveats. First, the conditioned PPO-GRU’s −29.2 bps result may reflect overfitting to Terra/LUNA’s specific depth-withdrawal timing rather than a fundamental RL

limitation, which additional cross-mechanism training pairs could resolve. Second, unlike reactive-simulator approaches such as Olby et al. [23], the PPO-GRU is trained on historical replay without market-impact feedback. In a thin-book stress regime this systematically miscalibrates entry sizing because the agent observes book depth as it was rather than as it would be after its own orders clear liquidity, which is a plausible contributor to the 919-trade overtrading pattern. Integrating the cross-mechanism protocol shown here into a reactive simulator is the most direct path to resolving the timing component of the oracle gap.

8 Conclusion

The 12× optical-to-executable gap establishes that stablecoin arbitrage benchmarks built on price labels measure the wrong quantity. The executable fraction of apparent opportunities is small, shrinks with notional size, and is invisible to every model trained on calm-market data. Cross-mechanism training on a structurally different stress event is the only empirically validated path to positive net returns, and the RL diagnostic provides a precise account of why: label density enables trading, but binary supervision over the profitable subset provides the directional accuracy that reward optimisation cannot replicate at low positive rates. The benchmark, labels, oracle implementation, and training scripts are released under CC-BY 4.0 (URL redacted for review) to support the next generation of execution-aware stress models.

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